

May 29, 2024

Name: Student number:

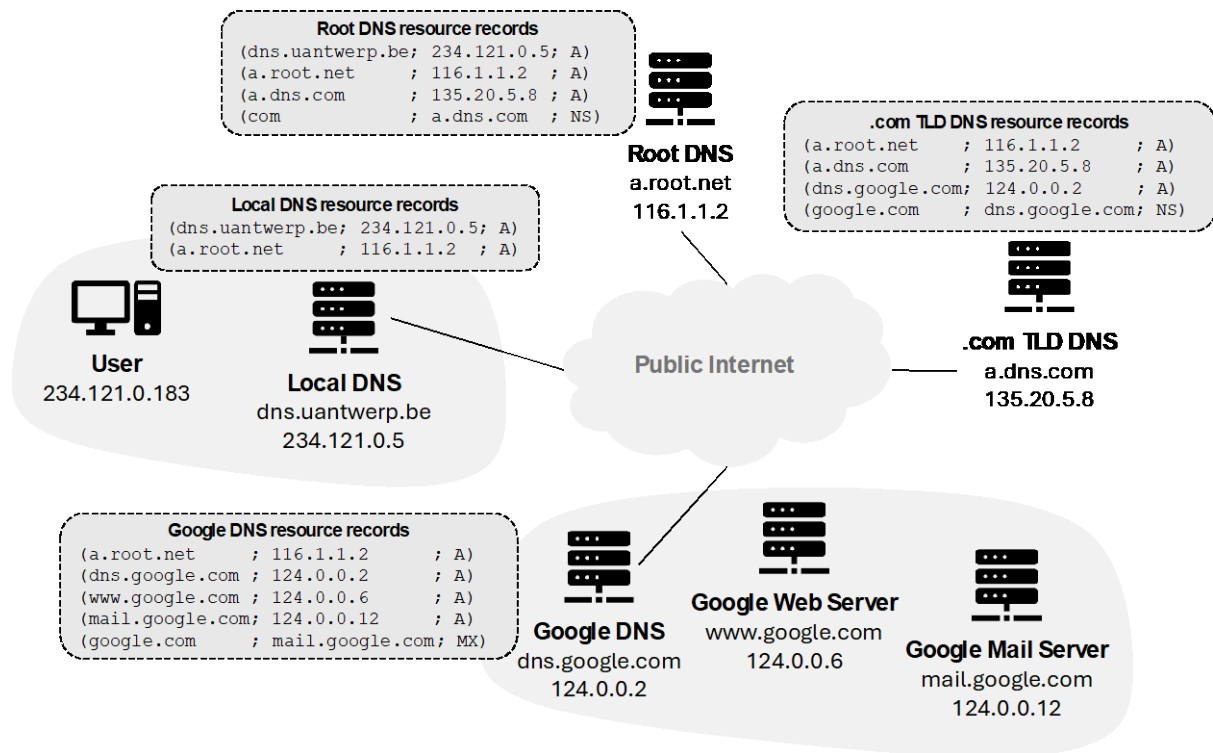
General remarks

- Fill out your name and student number at the top of each page.
- Place your student card visibly on your desk.
- You may bring 1 double-sided A4 sheet with notes.
- No other course material, notes, or book may be consulted during the exam.
- The use of a cell phone or any other means of communication is strictly forbidden.
- Always motivate your answer. Show how you came to your results.
- The expected size of the answer is in line with the size of the provided answer box.
- If your answer does not fit within the box, ask for an extra piece of paper. Clearly mark that your answer continues elsewhere.
- Plan your time well, you have **3 hours**.
- You can use a calculator.
- You are free to provide answers either in English or in Dutch.

Question 1: Domain Name System (DNS)

Consider the network as pictured below. It consists of a user (IP address: 234.121.0.183), a webserver (host name: www.google.com, IP address: 124.0.0.6), a mail server (host name: mail.google.com, IP address: 124.0.0.12), and 4 DNS servers: a local DNS (host name: dns.uantwerp.be, IP address: 234.121.0.5), a Google DNS (host name: dns.google.com, IP address: 124.0.0.2), a .com top level domain (TLD) DNS (host name: a.dns.com, IP address: 135.20.5.8), and a root DNS (host name: a.root.net, IP address: 116.1.1.2).

You may assume that the user knows the hostname and IP address of its local DNS (dns.uantwerp.be) through DHCP. The Google DNS server (dns.google.com) is the authoritative DNS server for the web server (www.google.com) and mail server (mail.google.com). The DNS resource records known to each DNS server are shown in a grey box next to each server in the figure, and structured as (name; value; type).



Name: Student number:

1. Assume the user (234.121.0.183) knows only the IP address and hostname of its local DNS server (234.121.0.5, dns.uantwerp.be), and the hostname of the web server (www.google.com). It now wants to set up an HTTP connection to the web server. List **all** the DNS requests and responses **in chronological order** that are exchanged in order for the user to find out the IP address of the Google web server assuming **iterative** DNS is used. For each request, mention the sender (by IP address), receiver (by IP address), DNS name, and DNS type. For each response, mention the sender, receiver, DNS name, DNS value, and DNS type.

Name: Student number:

2. List all requests and responses that are exchanged in chronological order in case **recursive** instead of iterative DNS is used by all DNS servers.

3. Now assume the user wants to send an email to `feedback@google.com`. Moreover, the local DNS server has cached the following resource records from resolving the webserver IP address: (`google.com`; `dns.google.com`; NS) and (`dns.google.com`; `124.0.0.2`). List all DNS requests and responses that are exchanged in chronological order in case iterative DNS is used, for the user to find out the IP address of the Google email server.

Question 2: Go-Back-N

1. Consider a sender A and receiver B that want to reliably communicate over an unreliable computer network. They make use of the Go-Back-3 ($N = 3$), and a sequence number of 2 bits. B receives 4 packets from A, with sequence numbers 0, 1, 2, and 0 (in that order). B sends back an acknowledgment for each of the packets immediately upon receiving them. What problem can occur in this scenario? Give an example.

2. Give two possible solutions to solve the identified problem. Use the same example as above to illustrate that it works.

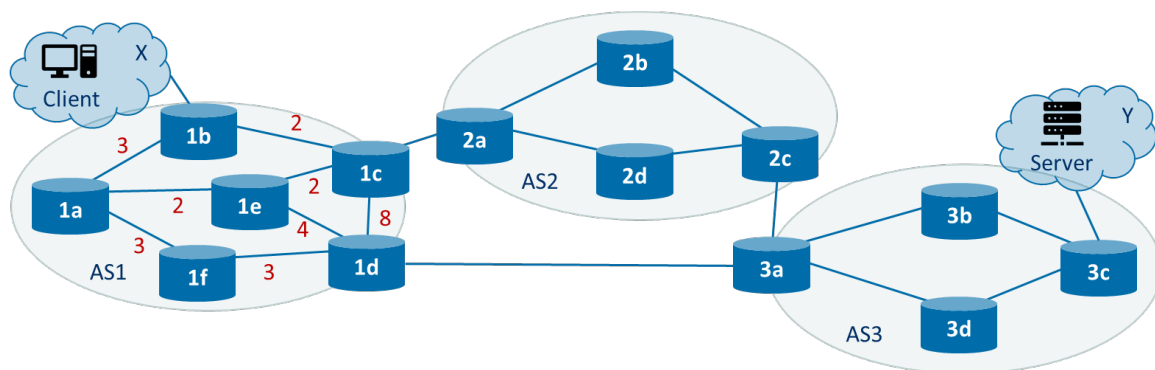
Name: Student number:

3. Draw a finite state machine for the general Go-Back-N **sender**. Be sure to include all necessary states, events, and actions.

4. Why does Go-Back-N require an estimation of the round-trip time (RTT)? What happens of the RTT is estimated too high or too low?

Question 3: Internet Routing

Consider the network scenario as shown in the figure below. It consists of 3 autonomous systems (labelled AS1, AS2, AS3), each consisting of several routers (labelled 1a, 1b, ..., 3c, 3d). The number (in red) next to each internal link in AS1 represents the OSPF link cost. Consider a client located in subnet X connected to router 1b in AS1 that wants to stream a movie using an HTTP Adaptive Streaming (HAS) based protocol from a Netflix server located in subnet Y connected to router 3c in AS3. Within each autonomous system OSPF routing is used, while BGP is used between them.



1. Explain, in detail, the process of how router 1b discovers the route to subnet Y using BGP.

Name: Student number:

2. Assuming the *local preference value* feature is not used, which route would be selected between the client and server by BGP? Explain why.

3. Which route within AS1 would OSPF select to route traffic between the client and server? Explain step by step how this route is determined.

Question 4: Multiple access protocol

Consider the Frequency Division Slotted ALOHA (FDS-ALOHA) multiple access protocol, which combines Frequency Division Multiple Access (FDMA) with Slotted ALOHA. In this scheme, the total available bandwidth is split into smaller equal-sized subcarriers. When a node wants to transmit, it selects a subcarrier uniformly at random, and then uses the rules of Slotted ALOHA to transmit the data. For each retransmission, the random subcarrier selection is performed again.

1. Consider an FDS-ALOHA scheme with $S=5$ subcarriers, and $N=100$ nodes. You may assume that each node always has a packet to transmit, and performs a transmission (or retransmission) with a probability $p=0.01$ during each time slot. You may further assume that transmissions in different subcarriers do not interfere with each other. What is the efficiency of FDS-ALOHA under these assumptions (you may round to 2 digits after comma)? Provide the different steps in your calculation and explain them.

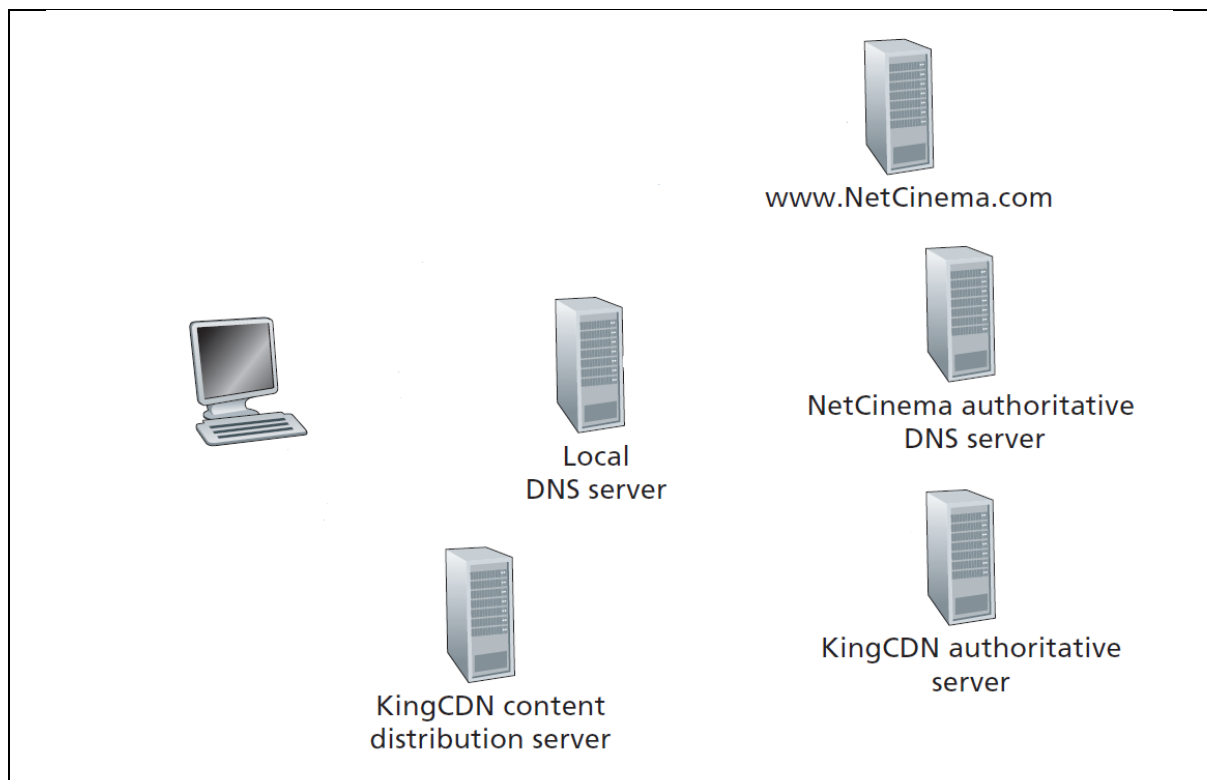
Name: Student number:

2. What would be the efficiency of regular Slotted ALOHA given the same parameters? Which of the two approaches is most efficient? Explain why.

Question 5: Content Delivery Networks (CDN)

A client wants to stream their favorite show over the Internet. Thus, they connect to a video streaming service (NetCinema) which uses a 3rd party CDN (KingCDN).

1. How does NetCinema deal with the client's request? In the figure below show the messages exchanged in order to start the streaming (using numbered arrows) and explain the different steps of the process.



Name: Student number:

2. Imagine now that the streaming service is Netflix. What are the main differences between the prior approach and the approach of Netflix?